



Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY ADVANCED TECHNICAL BOOTCAMP · 40 HOURS

Deep Learning & Computer Vision

Neural networks, CNNs, Transformers, and production vision systems

5

DAYS

40

HOURS

20+

HANDS-ON LABS

6

DL PROJECTS

Course Overview

This 5-day advanced bootcamp covers deep learning theory and practice — from neural network fundamentals through CNNs, RNNs, and Transformer architectures to production computer vision systems. Participants implement 6 end-to-end projects using TensorFlow, PyTorch, and Hugging Face across image classification, object detection, and generative vision tasks.

From math to production. This bootcamp bridges the gap between understanding neural network theory and deploying real vision systems. You'll implement architectures from scratch, fine-tune pretrained models, and deploy inference pipelines — covering the full stack that enterprise AI teams need for banking fraud detection, manufacturing quality control, healthcare imaging, and document intelligence.

Who Is This For?



ML Engineers

Engineers building and deploying deep learning models



Researchers

Data scientists exploring advanced neural architectures



CV Engineers

Engineers building image and video processing systems



AI Architects

Architects designing enterprise vision solutions

Learning Outcomes

- 1 Build neural networks from scratch** — implement perceptrons, activation functions, backpropagation, and optimization using NumPy and then TensorFlow/PyTorch
- 2 Master CNNs for computer vision** — implement and fine-tune architectures like ResNet, VGG, EfficientNet for classification, detection, and segmentation
- 3 Apply sequence models** — build RNNs, LSTMs, and GRUs for time-series forecasting, text generation, and sequential data processing
- 4 Understand Transformer architectures** — attention mechanisms, BERT, Vision Transformers (ViT), and their applications in NLP and vision
- 5 Build production vision pipelines** — object detection (YOLO), image segmentation, OCR, and document intelligence using OpenCV and Azure AI Vision
- 6 Deploy deep learning models** — ONNX export, TensorRT optimization, containerized inference with FastAPI, and cloud deployment

Prerequisites

Python programming (intermediate), basic machine learning knowledge (Bootcamp 4 or equivalent), linear algebra fundamentals (matrices, vectors). GPU-enabled environment provided during labs.

5-Day Curriculum

Interactive format with live coding, hands-on labs, and project-based learning. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Neural Network Foundations — From Math to Code

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A

Deep Learning Fundamentals

- Neurons, activation functions (ReLU, sigmoid, softmax)
- Forward propagation and loss functions
- Backpropagation and gradient descent (SGD, Adam, RMSprop)
- Lab: Build a neural network from scratch using NumPy
- Regularization: dropout, batch normalization, L1/L2

SESSION B

TensorFlow & PyTorch Essentials

- TensorFlow 2.x: tensors, layers, Sequential and Functional APIs
- PyTorch: tensors, autograd, nn.Module, DataLoaders
- Training loops, callbacks, checkpointing, TensorBoard
- Lab: Build and train an MNIST classifier with TensorFlow
- Lab: Build the same classifier with PyTorch — compare workflows

02

Convolutional Neural Networks — Image Intelligence

LIVE CODING + HANDS-ON LABS + ARCHITECTURE WORKSHOP

SESSION A

CNN Architecture & Image Classification

- Convolution, pooling, stride, padding — visual intuition
- Classic architectures: LeNet, AlexNet, VGG, ResNet
- Modern architectures: EfficientNet, MobileNet
- Lab: Build a custom CNN for medical image classification
- Lab: Transfer learning with ResNet on a custom dataset

SESSION B

Advanced CNN Techniques

- Data augmentation strategies for small datasets
- Fine-tuning pretrained models on domain-specific data
- Grad-CAM: visualizing what CNNs learn
- Lab: Fine-tune EfficientNet for manufacturing defect detection
- Lab: Build an image similarity search engine

03

Sequence Models & Transformers

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A**RNNs, LSTMs & Sequence Processing**

- Recurrent Neural Networks: architecture and limitations
- LSTMs and GRUs: solving the vanishing gradient problem
- Bidirectional and stacked RNN architectures
- Lab: Time-series forecasting for stock price prediction
- Lab: Text generation using character-level LSTM

SESSION B**Transformer Architecture & Attention**

- Self-attention mechanism — intuition and mathematics
- Multi-head attention, positional encoding, encoder-decoder
- BERT, GPT architecture overview (for understanding, not training)
- Vision Transformers (ViT): applying Transformers to images
- Lab: Fine-tune a Hugging Face BERT model for text classification

04

Computer Vision — Detection, Segmentation & OCR

LIVE CODING + HANDS-ON LABS + INDUSTRY PROJECTS

SESSION A**Object Detection & Segmentation**

- Object detection architectures: YOLO (v8/v11), SSD, Faster R-CNN
- Instance segmentation with Mask R-CNN
- Semantic segmentation with U-Net
- Lab: Build a real-time object detection system with YOLOv8
- Lab: Vehicle and pedestrian detection for traffic analysis

SESSION B**OCR, Document AI & Azure Vision**

- OpenCV fundamentals: image processing, contours, transformations
- OCR pipelines: Tesseract, EasyOCR, Azure Document Intelligence
- Azure AI Vision: image analysis, face detection, custom vision
- Lab: Build a cheque/ID document extraction pipeline
- Lab: Invoice processing with Azure Document Intelligence

05

Generative Vision, Deployment & Capstone

CAPSTONE PROJECT + DEPLOYMENT LAB + PRESENTATIONS

SESSION A**Generative Models & Advanced Vision**

- Autoencoders: vanilla, variational (VAE), for anomaly detection
- GANs: generator-discriminator architecture, DCGAN
- Diffusion models overview (Stable Diffusion, DALL-E concepts)
- Lab: Build a VAE for anomaly detection in manufacturing
- Lab: Image captioning with vision-language models

SESSION B**Model Deployment & Capstone**

- ONNX export and model optimization (TensorRT, quantization)
- Serving vision models with FastAPI and Docker
- Edge deployment considerations: mobile, IoT, embedded
- Capstone: End-to-end vision project — team presentations
- Certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools used hands-on with live coding and lab exercises

TensorFlow 2.x

PyTorch

Keras

Hugging Face

OpenCV

YOLOv8

ONNX

TensorBoard

Azure AI Vision

Azure Document Intelligence

FastAPI

Docker

NumPy

Matplotlib

Jupyter / Colab

What You Take Home

6 DL Projects

Complete notebooks — MNIST classifier, medical imaging CNN, YOLO detector, OCR pipeline, anomaly VAE, image captioning

Architecture Reference

Visual guides to CNN, RNN, LSTM, Transformer, and Vision Transformer architectures with implementation notes

Vision Pipeline Kit

Production-ready OpenCV + YOLO + OCR pipeline templates for document processing and quality inspection

Model Zoo Cheat Sheet

Quick reference for pretrained model selection — accuracy, speed, size tradeoffs for ResNet, EfficientNet, YOLO, ViT

Certification of Completion



Gennoor Tech — Certificate of Completion

Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Deep Learning & Computer Vision". Digitally signed by the Founder & Director. Validates proficiency in neural networks, CNNs, Transformers, object detection, OCR, and production vision deployment.

Recommended Next Certifications

Microsoft AI-103

Azure AI App & Agent Developer Associate

Microsoft AI-300

Machine Learning Operations (MLOps) Engineer Associate

TensorFlow Developer

Google TensorFlow Certificate

AWS ML-S-C01

AWS Machine Learning Specialty

Designed & Delivered By

Jalal Ahmed Khan

Founder & Director, Gennoor Tech Private Limited

Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of experience. Delivered deep learning and computer vision programs for enterprise teams. Built production document intelligence and OCR pipelines for banking workflows. Open-source CV projects on GitHub including multimodal RAG with Florence embeddings.

16+ active Microsoft certifications including AI-103, AI-300, AZ-400, PL-400, and GitHub certifications. Trained teams at Microsoft, IBM, Boeing, Saudi Aramco, and 80+ organizations across 6+ countries.

M.E. Mechanical Engineering, Mumbai University.

Delivery Options



On-Site — customized at your office
UK



Virtual Live — interactive via Teams/Zoom



International — GCC, Africa, APAC,



Ready to See Through the Eyes of AI?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

 www.gennoor.com



 +91 9326 352 241

 linkedin.com/company/gennoor-tech-private-limited

 YouTube — @GennoorTech

Genn